Airline Passenger Satisfaction Analysis

Dataset description: The commonly used Kaggle Airline Passenger Satisfaction training dataset contains passenger travel details, service ratings, delay information, and a satisfaction label. This example uses the file **train.csv** and the column **satisfaction**.

Python program

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset (keep train.csv in the same folder)
df = pd.read_csv("train.csv")

print("Dataset Shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head())
print("\nSatisfaction Counts:")
print(df["satisfaction"].value_counts())

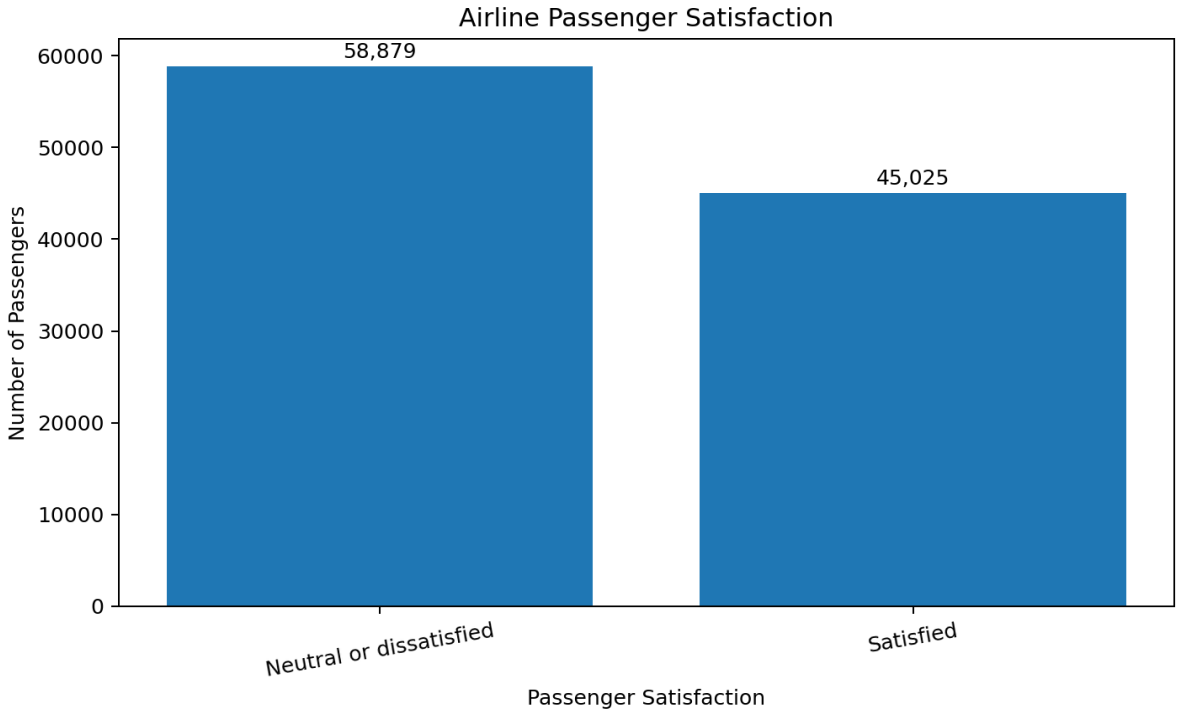
# Plot satisfaction distribution
df["satisfaction"].value_counts().plot(kind="bar")
plt.title("Airline Passenger Satisfaction")
plt.xlabel("Satisfaction")
plt.ylabel("Number of Passengers")
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig("airline_satisfaction_graph.png")
plt.show()
```

Example output

For the commonly used Kaggle training file, the dataset shape is typically (103904, 25). The satisfaction counts are shown below. Your output may differ if you use a different version of the dataset.

Satisfaction	Count
Neutral or dissatisfied	58,879
Satisfied	45,025

Graph



Conclusion: In this dataset, the neutral-or-dissatisfied group is larger than the satisfied group. Airlines can study service ratings, seat comfort, boarding experience, and delays to identify areas for improvement.

Note: The graph and counts above use the standard Kaggle training dataset's commonly reported class counts. Run the program on your own train.csv file to generate results from your exact dataset.